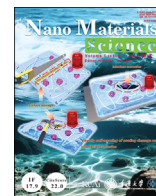


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The integration of microelectronic and photonic circuits on a single silicon chip for high-speed and low-power optoelectronic technology



Rajeev Gupta^{a,*}, Ajay Kumar^b, Manoj Kumar^c, Rajesh Singh^{d,e}, Anita Gehlot^{d,e}, Purnendu Shekhar Pandey^f, Neha Yadav^g, Kailash Pandey^a, Ashish Yadav^{g,**}, Neha Gupta^h, Ranjeet Brajpuriya^a, Shalendra Kumar^a, Ajay Singh Vermaⁱ, Tanuj Kumar^j, Yongling Wu^{g,***}, Zheng Hongyu^g, Abhijit Biswas^k, Ajay Mittal^l, Aniruddha Mondal^m, Romanov Oleksandr Ivanovichⁿ

^a Department of Physics, School of Engineering, University of Petroleum & Energy Studies, Dehradun-248007, Uttarakhand, India

^b Jaypee Institute of Information Technology, Noida, Uttar Pradesh, India

^c MLR Institute of Technology, Hyderabad, India

^d Uttaranchal Institute of Technology, Uttaranchal University, Dehradun, 248007, India

^e Department of Project Management, Universidad Internacional Iberoamericana, Campeche, 24560, CP, Mexico

^f Department of Electronics and Communication Engineering, GL Bajaj Institute of Technology and Management, Affiliated to AKTU, Greater Noida- 201306, U.P, India

^g Center for Advanced Laser Manufacturing (CALM), Shandong University of Technology, Zibo, 255000, PR China

^h Applied Science and Humanity Department, Dronacharya Group of Institutions, Greater Noida, India

ⁱ Division of Research and Innovation, School of Applied and Life Sciences, Uttaranchal University, Dehradun, Uttarakhand, 248007, India

^j Department of Nanoscience & Materials, Central University of Jammu, Jammu, 181143, India

^k USA Prime Biotech LLC, 1330 NW 6th Street, Suite A2, Gainesville, FL 32601, USA

^l Energy Cluster, School of Engineering, University of Petroleum & Energy Studies, Dehradun, 248007, Uttarakhand, India

^m Department of Physics, National Institute of Technology Durgapur, Durgapur, 713209, West Bengal, India

ⁿ Institute of Telecommunication Systems, National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute", Kyiv, Ukraine

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ABSTRACT

The combining microelectronic devices and associated technologies onto a single silicon chip poses a substantial challenge. However, in recent years, the area of silicon photonics has experienced remarkable advancements and notable leaps in performance. The performance of silicon on insulator (SOI) based photonic devices, such as fast silicon optical modulators, photonic transceivers, optical filters, etc., have been discussed. This would be a step forward in creating standalone silicon photonic devices, strengthening the possibility of single on-chip nano-photonic integrated circuits. Suppose an integrated silicon photonic chip is designed and fabricated. In that case, it might drastically modify these combined photonic component costs, power consumption, and size, bringing substantial, perhaps revolutionary, changes to the next-generation communications sector. Yet, the monolithic integration of photonic and electrical circuitry is a significant technological difficulty. A complicated set of factors must be carefully considered to determine which application will have the best chance of success employing silicon-based integrated product solutions. The processing limitations connected to the current process flow, the process generation (sometimes referred to as lithography node generation), and packaging requirements are a few of these factors to consider. This review highlights recent developments in integrated silicon photonic devices and their proven applications, including but not limited to photonic waveguides, photonic amplifiers and filters, on-chip photonic transceivers, and the state-of-the-art of silicon photonic in multidimensional quantum systems. The investigated devices aim to expedite the transfer of silicon photonics from academia to industry by opening the next phase in on-chip silicon photonics and enabling the application of silicon photonic-based devices in various optical systems.

* Corresponding author.

** Corresponding author.

*** Corresponding author.

E-mail addresses: rajeev@ddn.upes.ac.in (R. Gupta), ashish@sdut.edu.cn (A. Yadav), wuyl2019@sina.com (Y. Wu).

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1. Introduction

The maturation of silicon photonics technology has raised expectations for its impact in creating vital new applications. These include ultra-fast optical communications, distance sensing in self-driving cars, photonic-enhanced computing, and quantum information processing. The unique combination of performance, high yield, and the ability to handle large volumes, facilitated by standardized manufacturing technologies, has contributed to the success of silicon photonics. Currently, thanks to standardized technology platforms in silicon photonics, foundries can access optimized components like low-loss optical paths, rapid modulators, adjustable tuning, swift germanium photodiodes, and highly efficient optical and electrical interfaces [1]. A silicon chip combining microelectronic and photonic circuits could provide unprecedented functions in high-speed computing, communications, and sensing [2]. Such an on-chip integration of microelectronics and photonics technologies could pave the way for significant breakthroughs in realizing high-speed, low-power consumption-based advanced optoelectronic technology [3]. Data centers, for example, would have high-performance computer processors and communications links that are low power, scalable, and consume low power when a large number of microelectronic and photonic circuits are integrated on a single chip [4, 5]. Fig. 1 shows a schematic illustration of the integration of various components on a silicon chip for next-generation optoelectronic technology applications [6].

Achieving the goal of integrating these two technologies on a single silicon chip is quite challenging. Photonic circuits typically employ silicon-on-insulator as a substrate, as opposed to bulk silicon for electronic circuits [7,8]. Computer processors, memory chips, communication chips, and image sensors are manufactured using complementary metal oxide semiconductor (CMOS) technology [9]. This technology can produce a large number of transistors and electronic circuits onto a single chip, as it is based on silicon [10–12]. Wafers made of silicon-on-insulator or bulk silicon substrates can be used for this technique [13,14]. Bulk silicon substitutes are the more popular choice due to their abundant supply chain and low cost [15]. On the other hand, thick wafers are employed in the fabrication process of silicon photonics. There are, however, limitations with thick wafers due to the limited supply chain, and they are also too expensive for many uses, such as computer memory [7,8].

Addressing the incompatibility difficulties between the substrate material platforms utilized in microelectronics and photonics poses a significant challenge [16–18]. Since bulk CMOS is not a semiconductor material with suitable optical properties, active and passive photonic functions cannot be implemented. This poses the most significant challenge in integrating photonic circuits into CMOS [3,8]. Integrating electronic and photonic components has been made possible through the use of CMOS manufacturing techniques and materials platforms [19,20]. That allows seamless integration of different components and does not affect their performances [7,8]. Furthermore, monolithic CMOS transistors provide time-multiplexing of electrical signals directly onto silicon photonic circuits to control silicon photonic circuits. This method has progressively interrogated photodetectors integrated into a Mach-Zehnder Interferometer (MZI) mesh. With increasing optical architecture complexity, on-chip multiplexing becomes increasingly appealing as external electronics become less connected to photonic circuits. This may allow optical integrated systems to be functionally scaled up further by combining multiplexers (MUX) and demultiplexers (DEMUX) [21].

Scientists have revealed the most current developments in microelectronics and photonics along with their applications as these fields continue to develop. They are now looking forward for opportunities to merge photonic and microelectronic circuits on a single silicon chip for a variety of industrial applications such as on-chip light source with high intensity and optical switches for high bandwidth communication in data centers [22,23]. Modern optical chips can be enhanced with electronic

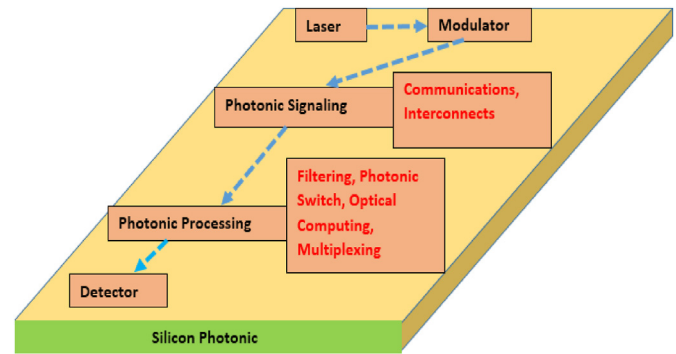


Fig. 1. A schematic illustration of integrating a large number of components on a single platform for advanced optoelectronic technology [6].

functionality by integrating transistors and circuits onto a silicon photonics platform without altering the usual fabrication method used by photonic foundries. This strategy maintains optical excellence and foundry interoperability while including on-chip electrical building blocks to enable optical capability at no additional expense. In a recent development, a transconductance amplifier and an analogue multiplexer for sequential readout of 16 photodetectors, spread in a large-scale design of MZIs, have been integrated to demonstrate the advantages of monolithic electronics [24]. In order to integrate photonic integrated circuits into well-designed optoelectronic systems, their packaging is essential and critical. Since photonic packaging is expensive and limited in throughput, it often poses a significant barrier to scalable deployment of PIC technologies [25]. Additionally, this field is quite recent and will soon revolutionize the global electronic industries [Fig. 2].

This figure is a significant asset for researchers, industry experts, and decision-makers, promoting collaboration and propelling progress in this rapidly developing field. In this review, authors have summarized the trends, patterns, and research gaps on microelectronic and photonic circuits by providing a comprehensive overview, analysis of existing research, and potential breakthroughs in these fields to improve quality of life by enabling new technologies and applications. In order to document the field's developments, trends, obstacles, and prospects, it is vital to have year-by-year publications on integrating photonic and electronic circuits onto a single silicon device. Integrating microelectronics with photonics has led to novel data center applications and capabilities, including high-speed optical interconnects and efficient data processing. For modern communication technologies like 5G and beyond, seamless integration of microelectronics and photonics is needed to provide more efficient and compact systems that reduce power consumption and latency. Considering this, researchers promote cutting-edge research that enables new technologies and applications to boost economic growth.

2. Recent development of emerging microelectronic and photonic circuits

Microelectronics, photonics, and other “high-technology” engineering are discussed in terms of some of the most prominent current developments.

2.1. Integrating photonic devices with state-of-the-art electronics on a bulk silicon microchip

Few strategies have been developed to address the restrictions associated with integrating photonic devices and advanced electronics [26–28]. Their methodology demonstrates considerable potential for achieving high-speed, energy-efficient optoelectronic technology. The researchers announced the initial achievement of producing photonic devices on a large silicon base. The device was constructed using a vast number of electronic components called transistors (Fig. 3) [7]. This

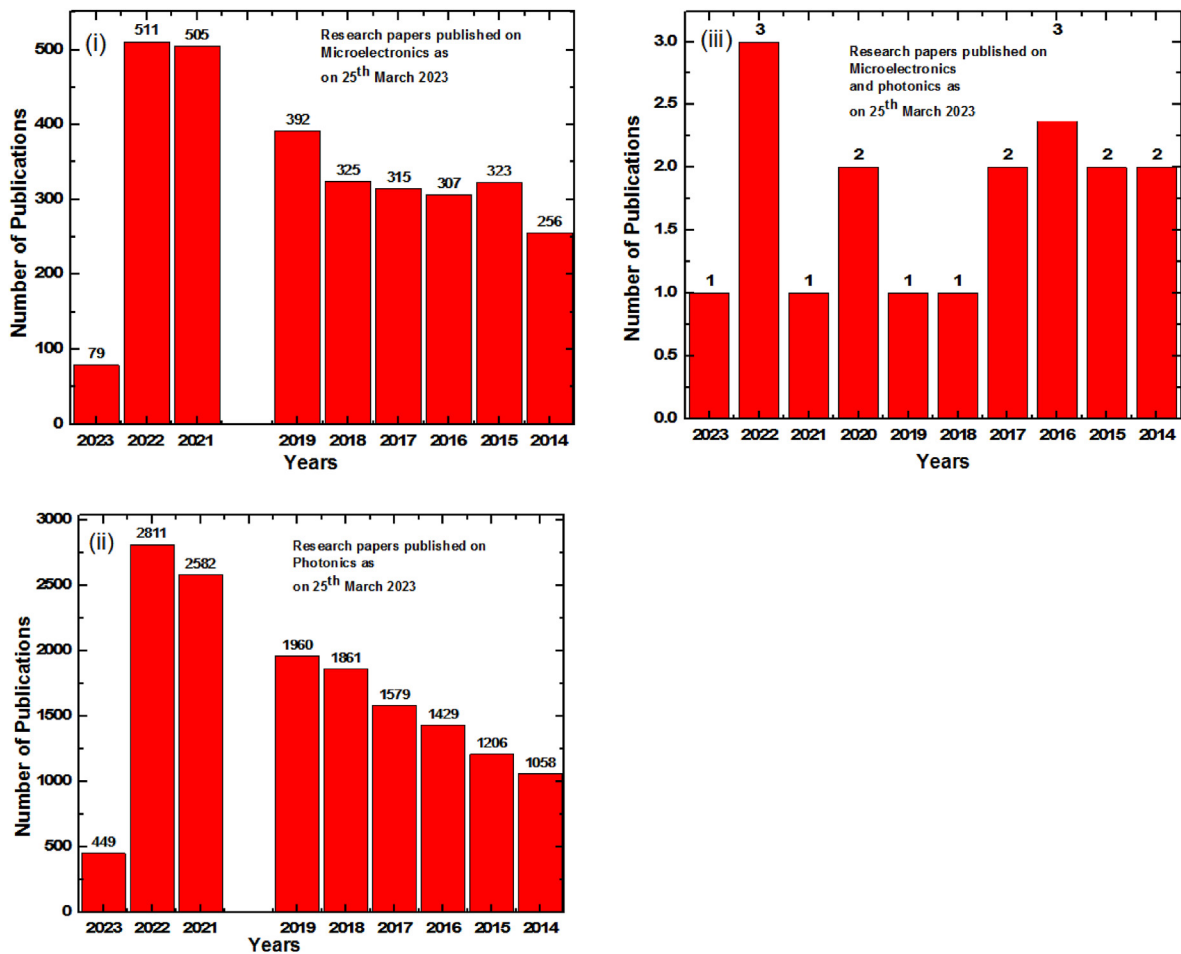


Fig. 2. Research paper published (i) on Microelectronics (ii) on Photonics (iii) on Integration of Microelectronic and Photonic Circuits (source: Web of Science).

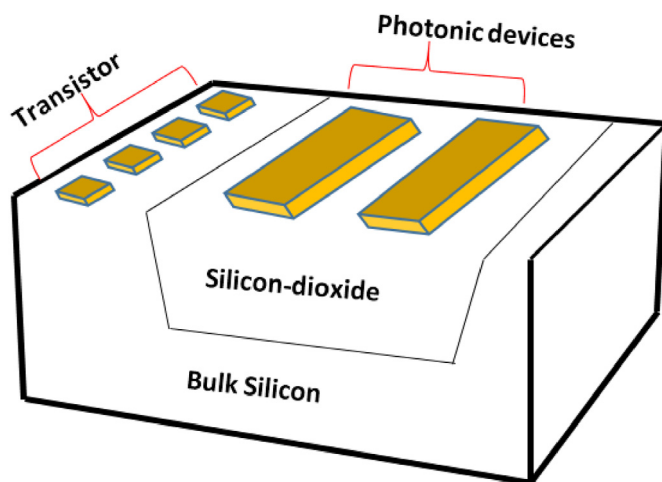


Fig. 3. A fabricated device showing optoelectronic integration that combines electronic and photonic devices on a single silicon microchip [7,8].

groundbreaking research facilitates the large-scale manufacturing of optoelectronic systems on microchips [8]. Photonics technology was integrated into CMOS chips by depositing polycrystalline silicon over silicon oxide islands (glass) [3,8]. Fig. 3 illustrates the surface of the silicon covered with thin layers of polycrystalline silicon. Afterwards, transistors, which are devices that use photons or electrons, were created from this layer in the silicon-on-insulator area and bulk silicon,

respectively [8].

An optical waveguide and resonator can be created, along with a high-speed optical modulator, using a single deposited layer [29–31]. Furthermore, they incorporated integrated high-speed optical transceivers into this constructed platform, which operates at a speed of 10 gigabits per second. This system consists of a vast number of transistors (as shown in Fig. 4) and is arranged on a single optical bus for the purpose of wavelength division multiplexing (WDM). This technology is important because it can meet the current need for fast and efficient connections in data centers and high-performance computing [31]. In addition, it may be inferred that separating the production of photonic devices from transistors could allow the integration strategy to attain the objectives of multi-chip systems, while maintaining the performance, cost effectiveness, and scalability of monolithic platforms [8,32,33].

2.2. The single-chip photonic transceiver on bulk silicon as an optical interconnect platform on the chip level

Silicon photonic integrated circuits, are designed with the particular purpose of facilitating the effective transmission and reception of light information [34]. There is a prevailing belief that the effective integration of photonic integrated circuits into prominent silicon electronic chips, namely as chip-level optical inputs/outputs (I/Os), has the potential to introduce transformative advancements in data computing and communications [35]. In order to achieve this objective, a pioneering approach to photonic integration has been established. This approach involves the implementation of a single-chip optical transceiver, which is built upon a monolithic-integrated vertical photonic I/O device. This device is equipped with a light source that is situated on bulk silicon, as

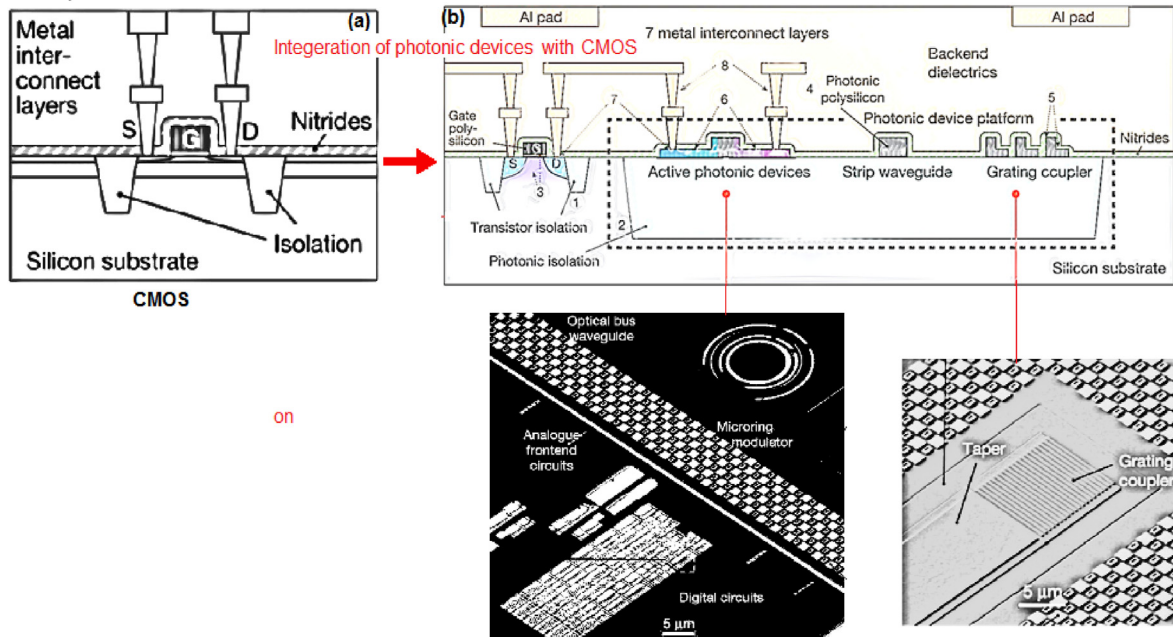


Fig. 4. The fabrication process steps for photonic integration with transistors [8].

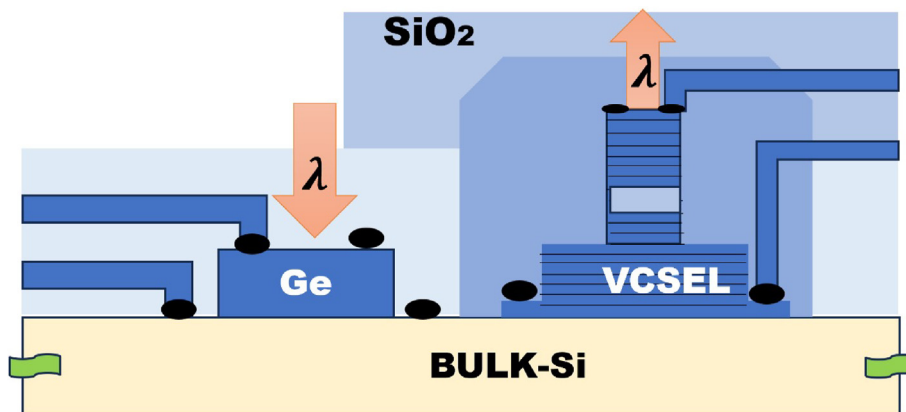


Fig. 5. A schematic diagram of the fabricated monolithically integrated complete photonic transceiver devices on bulk silicon wafer [35].

depicted in Fig. 5. Researchers have been investigating the feasibility of integrating monolithically integrated vertical-illumination Ge-on-Si photo-detectors on a single-chip photonic transceiver. Additionally, the configuration encompasses a vertical-cavity surface-emitting laser (VCSELs)-on-Si, which operates at speeds of up to 50 Gb/s and 20 Gb/s on the same bulk-silicon substrate [36,37]. The researchers presented a prototype that successfully demonstrated chip-level optical interconnects operating at a low power of 20 Gb/s. These interconnects were established between chips that were produced, with a wavelength of approximately 850 nm. There are several applications for this methodology for implementing electronic-photonic integration, including high-performance computers (HPCs), CPU-memory interfaces, hybrid memory cubes, local area networks (LANs), storage area networks (SANs), data centers, and network applications [35].

2.3. Multidimensional quantum systems: on-chip generation, manipulation and analysis of large-scale integrated quantum photonic technologies

The development of optical on-chip patterning and coupling circuits in single and entangled photon sources holds significant importance as it serves as the fundamental framework for an integrated quantum optics

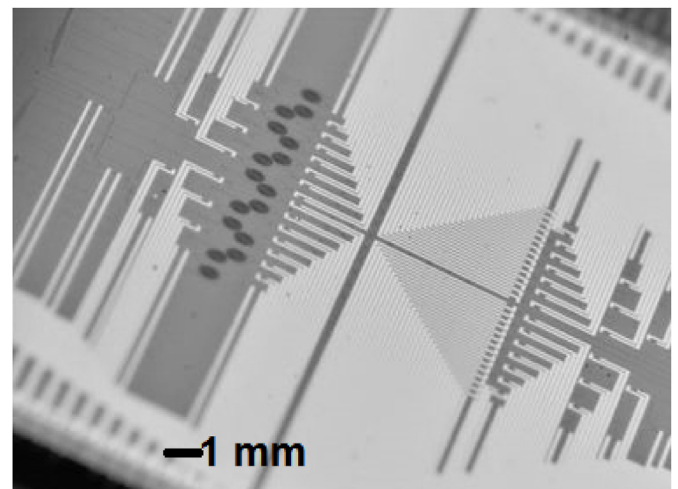


Fig. 6. A quantum silicon chip device displaying silicon waveguides with gold wires that are used to control phase shifters electronically [38].

platform. Researchers have investigated the potential of silicon photonics quantum technologies for facilitating the complete on-chip production, strategy, and consideration of multidimensional quantum systems. The research presented a significant advancement by successfully integrating over 550 quantum optical components and 16 photon sources onto a cutting-edge silicon chip, as illustrated in Fig. 6. The complexity attained by the integrated device allows for the universal production, manipulate, and investigation of multidimensional entanglement. This development signifies a notable advancement in the field of large-scale quantum photonic technology [38]. In prospective scenarios, the use of UV immersion lithography with a depth of 300 nm holds the prospective to enhance the implementation, yield, and scalability of components. This advancement could enable the integration of devices with a substantial number of components, reaching up to 100 000.

2.4. Ultra-compact silicon photonic filters: a high-performance technology

As silicon photonics develops into a wide range of applications, including optical connectivity, nonlinear optics, LiDAR, and quantum optics, it is experiencing an exponential development [39]. There are a wide variety of optical systems that use silicon photonic filters to modify the spectrum [40,41]. The silicon-on-insulator platform allows for the production of photonic integrated circuits, which facilitates the expansion of the CMOS infrastructure [42–44]. Ultra-compact silicon photonic filters characterized by high extinction ratios and box-like responses can be realized through the integration of tenth-order adiabatic elliptical-micro-rings (AEM) with bent directional couplers [40]. The active electro-optical materials incorporate waveguides that are situated in the non-coupling regions. These waveguides possess a wide structure and exhibit a modest bending radius, as reported in Refs. [45,46]. On the other hand, the waveguides situated in the coupling zones exhibit constriction and possess a significant bending radius [47,48]. The experimental results presented in Ref. [40] demonstrate the successful implementation of silicon photonic filters utilizing tenth-order all-electrical multiplexers, exhibiting a notable free spectral range of 37 nm [40]. The manufacturing process involved the utilization of CMOS-compatible deep-ultraviolet lithography, which employs radiation with a wavelength of 248 nm to print minute features measuring 250 nm. The present devices exhibit exceptional characteristics, showcasing minimal additional losses of 1 dB and strong extinction ratios above 60 dB [49]. Significantly, the currently available silicon photonic filters provide effective performance in a fully passive manner, without requiring any temperature control for calibration [50,51]. Integrated photonics has emerged as a prominent field, garnering substantial attention in recent years [52–54]. The production of compact optical filters that exhibit excellent performance characteristics, such as a high extinction ratio (ER), minimal excess loss, and a box-like response, is of utmost importance [52,55]. In order to mitigate the need for pumps in silicon quantum photonics, it is possible to employ an optical filter with a ER value, as suggested in Ref. [56]. Various photonic filters have demonstrated good outcomes through the utilization of MZIs, Bragg gratings, micro-ring resonators (MRRs), and their cascaded architectures [57]. Among the several alternatives available, high-order MRRs have emerged as particularly attractive due to their compact size and versatile functionality. High-order MRRs utilizing silicon photonic waveguides with ultra-high refractive index contrast face significant challenges due to fabrication imperfections, which produce random nanometer-scale dimensional variations that greatly affect the effective index. The consequence of this is a notable discrepancy in the resonance wavelength for the MRRs. The use of individual thermal tuning for each component of the micro-ring is a commonly employed approach for achieving wavelength alignment [58]. The aforementioned constraint prompted the development of this particular technique. Regrettably, this leads to a notable escalation in the energy consumption of the device, accompanied by an augmentation in both its physical dimensions and the intricacy of its electronic components. A further approach involves the utilization of

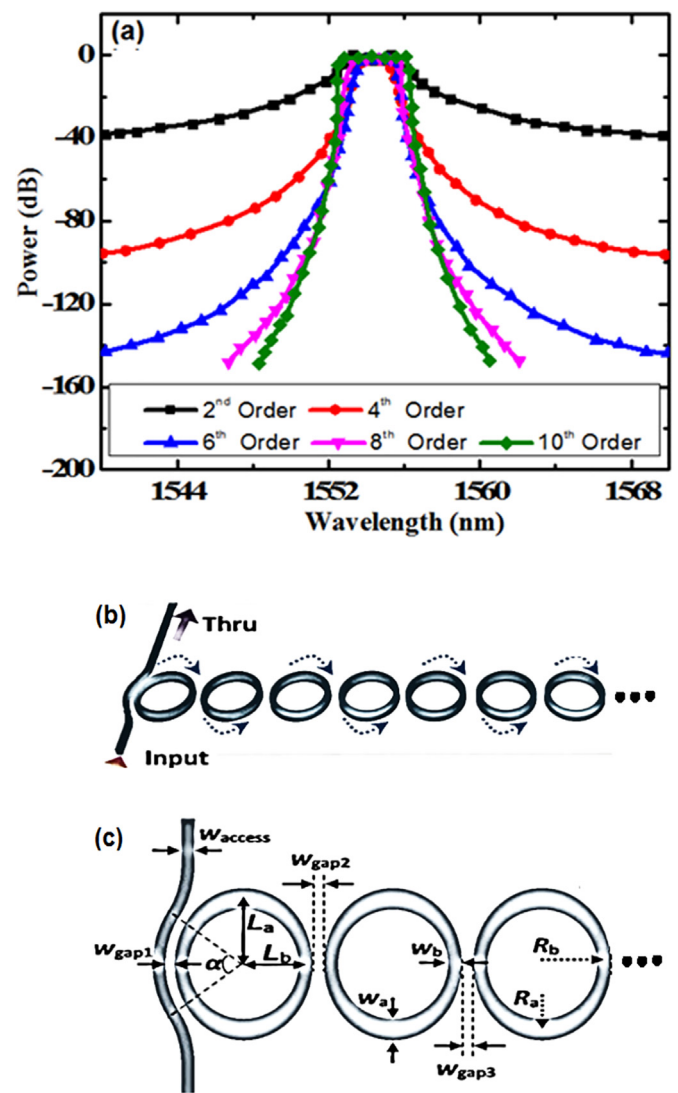


Fig. 7. An image of the silicon photonic filter with tenth-order AEMs (a) Shows a box-like spectral response with high ERs and high roll-off rates (b) Shows the three-dimensional and partial views (c) Shows the main parameters [44].

contemporary processing and manufacturing methodologies, shown by the implementation of 193 nm ArF-immersion lithography [59]. Nevertheless, the majority of silicon photonic foundries still lack the necessary technologically sophisticated equipment [60].

Using tenth-order AEMs, we can develop a silicon photonic filter with a 37-nm FSR, capable of achieving high optical efficiency. The aforementioned AEMs are able to provide the desired level of coupling strength and compact size while minimizing the loss of signal propagation by the adiabatic adjustment of the radius and width of the ring waveguide [40]. Previously, E-beam (electron-beam) lithography techniques have been employed to construct silicon photonic filters of superior quality, utilizing second-, third-, and fourth-order avalanche enhancement mechanisms. Nevertheless, the performance gains achieved by expanding the order beyond 10 are not substantial. In general, an increased number of cascaded rings tends to yield a spectral response that resembles a box shape, characterized by bigger extinction ratios (ERs) and higher roll-off rates, depicted in Fig. 7a. Consequently, silicon photonic filters based on the tenth-order anisotropic effective medium theory have become widely adopted in the field. The alignment of wavelengths is crucial for the optimal functioning of these 10 cascaded AEMs. Modern tenth-order AEM photonic filters on silicon may be effectively manufactured using commonly available and cost-effective

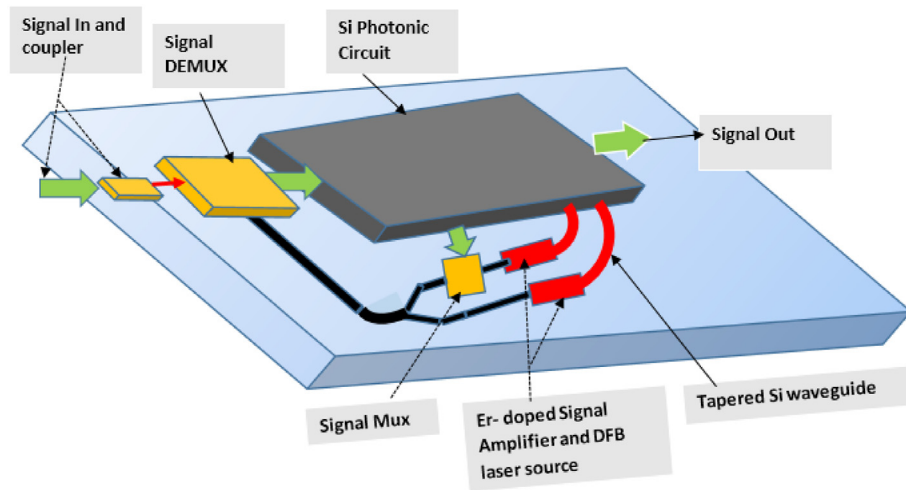


Fig. 8. Schematic configurations of compact erbium-doped based amplifiers with silicon-on-insulator waveguides [63].

foundry processes, resulting in exceptional performance. The current highest recorded value for the tenth-order MRR is 37 nm, while achieving an ER of 60 dB is crucial for the successful implementation of AEM photonic filters. Nevertheless, the dimensions of an individual AEM are merely $5.6 \times 4.6 \mu\text{m}^2$.

The current footprint of the AEM is the most miniature among all high-order MRRs that have been recorded, which significantly contributes to the reduction of the inherent variability in the resonance wavelengths of these AEMs. The enhanced separation between the AEMs and the extended waveguide additionally contributes to the enhanced robustness of the fabrication. The potential of on-chip high-ER photonic filtering is evident when considering the advantages associated with manufacturing and integration with CMOS technology. The Fig. 7b shows the three-dimensional (3D) representation of the AEM photonic filter with tenth-order. As a result, there is an adiabatic modification to the diameter and radius of the AEM ring waveguide. In regions where coupling does not occur, the waveguides are expanded in width and exhibit a reduced bending radius [61]. The increased width of the waveguide enhances the ring's tolerance towards fluctuations in width due to the reduced susceptibility of the effective index to such alterations. Even when fabrication uses a relatively large gap width, bent directional couplers (DCs) improve coupling efficiency by enhancing direct coupling between the access waveguide and the adjacent AEM. Furthermore, the coupling zones exhibit narrow waveguides that possess a maximum bend radius. Hence, the gaps in the coupling regions may exhibit significant dimensions, so aiding the thorough filling of the upper silica coating. The utilization of advanced electromagnetic materials (AEM) and bent coupling structures enables the creation of miniature resonators that exhibit minimal additional losses and satisfactory coupling, even in scenarios where 10 AEMs are interconnected. The reduction in structural complexity and footprints is achieved through the minimization of phase error resulting from manufacturing flaws. According to Ref. [62], it is possible that the requirement to individually tune and calibrate each resonator could be eliminated.

Fig. 7c shows an SOI waveguide with a top silicon layer of 220 nm and an oxide buffer layer of 2 m thick. As reported, the operating wavelength is approximately 1550 nm, with silicon and silica having refractive indices of 3.476 and 1.455, respectively [49]. The devices were specifically engineered to accommodate transverse-electric (TE) polarization. The primary factors influencing the design of the AEMs are the half lengths (L_a and L_b) and the widths (W_a and W_b) at the major/minor axes, as shown in Fig. 7c. The radius and width of the AEMs are modified in an adiabatic manner, leading to compact dimensions that minimize propagation loss for the micro-resonators while ensuring sufficient coupling.

2.5. An integrated erbium-doped amplifier with silicon-on-insulator waveguide fabricated monolithically

It has been demonstrated how $\text{Al}_2\text{O}_3:\text{Er}^{3+}$ amplifier technology can be integrated monolithically with passive silicon-on-insulator waveguide technology [63]. A significant increase in signal strength of greater than 7 dB is achieved at a wavelength of 1533 nm. The integration of several amplifiers and laser sections into parallel configurations with silicon on a chip can be achieved by a straightforward wafer-scale manufacturing procedure. This method involves the process of reactive co-sputtering, trailed by subsequent reactive ion etching [64]. In the past decade, passive integrated optics technology has undergone significant advancements, resulting in its adoption as a robust photonics technology with wide application [65]. The utilization of silicon waveguides is advantageous due to their high refractive index contrast, which enables the construction of compact devices and densely integrated optical circuits. While silicon waveguides tend to exhibit higher propagation losses (ranging between 0.9 and 3 dB/cm) in comparison with Si-compatible dielectrics with 0.1 dB/cm, it is worth noting that silicon waveguides have typically higher propagation losses. Silicon photonics enables the realization of low-loss bends with radius as tiny as 5 μm . Moreover, the popularity of this technology stems from its compatibility with CMOS and the presence of a well-established processing infrastructure [66]. In addition to passive functionality, light detection and integrated modulation approaches are discussed. The indirect bandgap in Si still causes inefficient light production, which is a key flaw in Si photonics.

Researchers have proposed using semiconductor optical amplifiers (SOA) or dielectric thin films doped with rare earth ions (RE) to generate light on silicon devices. Both systems include unique advantages and downsides [67]. As opposed to the former method, which involves hybrid integration techniques, like die-to-wafer bonding, the latter allows direct wafer deposition onto silicon substrates. Nonetheless, the utilization of electrical pumping in semiconductor optical amplifiers (SOAs) presents a notable benefit when compared to the optical pumping technique employed in RE ions. Researchers have previously proposed using electrically pumped rare-earth-doped dielectrics in slotted silicon waveguides. However, empirical validation of this concept is yet to be conducted. The gain per unit length of SOAs is several hundred dB/cm, which is significantly higher compared to the few dB/cm gain per unit length exhibited by dielectric counterparts [68]. Nevertheless, there are certain applications in which dielectrics doped with RE elements are considered more favorable compared to III-V semiconductors. A recent achievement was the successful amplification of data rates up to 170 Gbit/s using an Er-doped waveguide amplifier on a Si wafer, without experiencing any negative impact on noise or patterning effects.

Furthermore, it should be noted that there exists a possibility for increased bit rates due to the amplification of a signal consisting of optical time division multiplexed and polarization multiplexed data at a high speed in Er-doped fibers, resulting in a transmission rate of 1.28 Tbit/s. On the contrary, the phenomenon of eye closure occurs during the execution of Sequential Optical Amplifiers (SOAs) in a gain regime that is either saturated or quasi-saturated. This can be attributed to the relatively short carrier lifetime associated with these SOAs, typically around 100 picoseconds [69]. The transient suppression and gain recovery can be induced by the carrier lifetime, which depends on the bit rate and sequence. Furthermore, while the typical linewidth of III-V distributed-feedback (DFB) lasers seen in the market falls within the range of 1–10 MHz, it has been demonstrated that laser line widths as narrow as 1.7 kHz may be attained in RE-doped materials using free-running single-longitudinal-mode DFB lasers. Finally, the influence of temperature on the performance of SOA is more significant compared to that on their dielectric counterparts. The phenomenon of mode hopping may occur due to the significant wavelength change observed in the spectrum as temperature increases. On the contrary, the issue of temperature stability can often be effectively mitigated through the utilization of dielectrics doped with rare earth elements [70]. This research focuses on the wafer-scale monolithic integration of active Er-doped dielectric waveguide technology with passive Si photonics. Additionally, this study presents a novel approach and presents the initial experimental findings [71]. In addition to the exposition of our specific device design, production specifications, and amplifier measurements, we also provide a comprehensive account of the integration concept and available technological alternatives. Fig. 8 illustrates the graphic depiction of the notion of integration. Passive circuitry, pump/signal multiplexing (MUX) and demultiplexing (DEMUX) sections, high-speed amplifiers, and narrow-linewidth lasers are combined with active Er-doped elements on a silicon-on-insulator (SOI) device. The active and passive functionalities are linked through silicon photonic wires that are gradually narrowed into the Er-doped dielectric channel waveguides. This design enables the efficient connection of the pump and indicator light with minimal loss. This coupler exhibits a substantial optical bandwidth and a high coupling efficiency, which is crucial for successfully utilizing active features that operate across vast gain bandwidths [72]. The utilization of a straight-forward two-step parallel approach can be employed to generate multiple duplicates of the active portions at any desired location on the device. This technique involves the deposition of the active thin film, followed by the execution of reactive ion etching on the channel waveguides. The technique of simultaneously powering several amplifier and laser sections can be accomplished by employing a single diode laser, dispersing its power across the integrated circuit, and injecting it into the Er-doped components, as long as there is a sufficient quantity of available pump power. To effectively implement this concept, it is essential to have waveguide layers that are of exceptional quality, possess significant gain across a wide range of wavelengths, use a reliable deposition technique that can be applied to silicon wafers on a large scale, and utilize an active waveguide material with a high refractive index (n) [73]. The latter enables the creation of miniature integrated gain devices with minuscule waveguide cross-sections. Due to the reduced requirement for pump power, a solitary diode laser can effectively power several active device parts on a chip.

The gain material employed is amorphous aluminum oxide (Al_2O_3) doped with erbium (Er). Al_2O_3 is an excellent medium for active ions because it has a broad range of transparency and a high solubility for RE-ions. Reactive co-sputtering enables convenient deposition on various substrates. It is feasible to create planar waveguides and channel waveguides using deposition and dry-etching techniques, respectively, that exhibit minimal losses of 0.1 dB/cm and 0.2 dB/cm, as reported in Ref. [74]. Furthermore, Al_2O_3 exhibits a much higher refractive index (1.65 at $\lambda = 1.55 \mu\text{m}$) in comparison to other commonly used dielectric hosts such as silica (1.45) and phosphate glass (1.55). Several high-speed amplifiers have been developed using the integration of multiple active

devices onto one chip, zero-loss optical power splitters have been developed, wavelength-selective ring lasers covering the entire telecom C-band have been developed, and ultra-narrow lasers have been developed. These advancements were made possible by utilizing Er concentrations of approximately 10^{20} cm^{-3} . Consequently, this technology fulfills all the necessary conditions for integration. Aluminum oxide doped with erbium ions ($\text{Al}_2\text{O}_3:\text{Er}^{3+}$) structures have been devised and incorporated into silicon photonics utilizing a distinctive approach. The Si- $\text{Al}_2\text{O}_3:\text{Er}^{3+}$ structure has demonstrated a signal enhancement of 7.2 dB at a wavelength of 1533 nm. The performance of these integrated structures, as demonstrated by their equivalence to traditional $\text{Al}_2\text{O}_3:\text{Er}^{3+}$ amplifier devices, confirms the feasibility of achieving internal net gain [75]. In an incident pump power of 260 mW, a silicon photonic device can achieve a net gain over 40 dB through optimization of silicon taper losses, fiber-to-chip coupling efficiency, and amplifier length and footprint. One such choice is to integrate active functions onto a passive SOI photonic substrate using $\text{Al}_2\text{O}_3:\text{Er}^{3+}$ in a monolithic wafer-scale manner. It is possible to create advanced optical characteristics by integrating high-speed amplifiers with arrays of narrow-line width DFB lasers [73].

2.6. CMOS-compatible implementation of optical networks-on-chip based on III-V/SOI technology

An integration technique compatible with CMOS technology, using III-V/silicon-on-insulator, has been developed for 200 mm wafer size. Using this technique, it is possible to integrate lasers, photodetectors, and wavelength selector circuits onto microdisks for optical networks-on-chip [76,77]. It is anticipated that the current electrical connections used for transferring data between CPU cores would replace optical interconnects and more advanced optical network-on-chip (ONoC) technologies [78–81]. These systems, which include small laser sources that are electrically injected, depend on silicon's ability to perform all photonic operations with electro-photonic integrated circuits [82–84]. Previous studies have successfully proved the use of III-V/silicon technology on 200 mm wafers for continuous wave operation microdisk lasers, photo-detectors, and wavelength selective circuits [76,85,86]. The CMOS-compatible integration technology employs III-V semiconductor materials for the fabrication of both lasers and photo-detectors. Simultaneously, SOI sheets are designed with patterns to create passive circuits [87–89]. Illustrated in Fig. 8, the above-IC technique refers to a specific procedure that modifies and adjusts the typical III-V material production phases to meet the requirements of a CMOS environment. This approach employs wafer-based transistors, metal interconnects, an InP micro source, a photonic layer, and an InGaAs photo-detector. The components utilized include metal interconnects, wafer transistors, an InP micro-source, a photonic layer, and an InGaAs photo-detector. In order to ensure compatibility, it was imperative to develop die-to-wafer bonding on a 200 mm scale and explore dry etching chemistry that could etch the III-V dice while maintaining the Si/ SiO_2 -based surface, thus replacing the typical gold-based connections commonly used in optoelectronics [76]. The die-to-wafer bonding technique is employed to enable the integration of III-V materials and circuits. A process SOI wafer is bonded with a patterned III-V die using a direct bonding $\text{SiO}_2/\text{SiO}_2$ construction [77]. By fully eliminating the InP substrate, a bonding yield of around 80 % has been achieved using this ongoing technique [90,91]. The photonic chips were created using a unique architecture that incorporates a layered structure of III-V materials. This architecture allows for the integration of laser and photo-detector heterostructures within the same epitaxial layer. In comparison with previous point-to-point connections, where certain dice are allocated to laser structures and others to photodetectors, this simplifies the optical communications integration process and reduces the physical size of optical communications [92–95].

The utilization of 248 nm deep ultraviolet (DUV) lithography is feasible because of the $1 \mu\text{m}$ overall epitaxy thickness, which effectively resolves focalization issues [96,97]. The optimization of the III-V

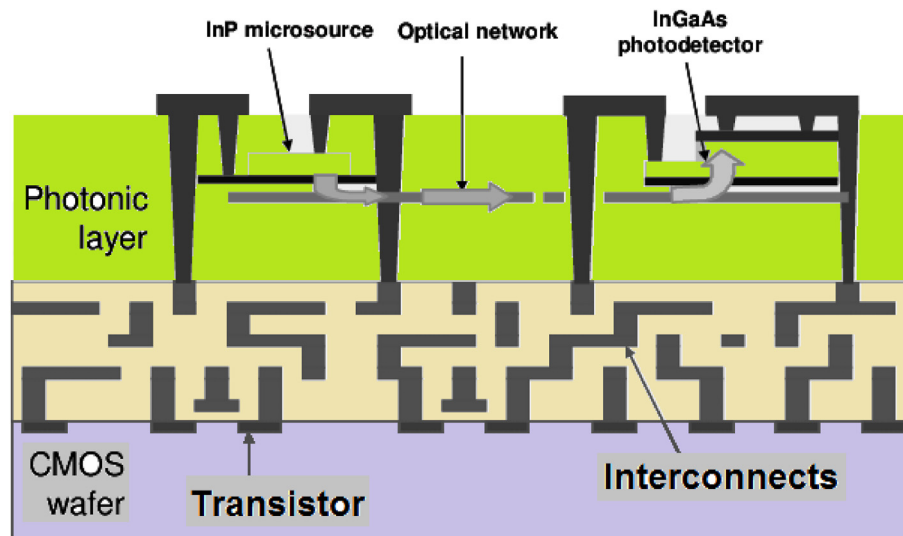


Fig. 9. Schematic of the optical networks-on-chip using CMOS [76].

technique is carried out on 200 mm wafers. In this procedure, an oxide isolation layer is deposited and the surface is metallized after wet reactive ion etching with $\text{CH}_4\text{-H}_2$ and deposition of an oxide isolation layer [98]. Fig. 8 illustrates an optical microscope image showing a 130 nm thick SiO_2 bonding layer that covers a waveguide, an III-V microdisk laser, and a photo-detector. Alternatively, a metal stack consisting of Ti/TiN/AlCu is positioned on top of the wafer. After lithography, a chlorine-based chemical is employed to perform dry-etching on the metal stack until it reaches the oxide isolation layer, which also functions as an etch-stop layer. By not subjecting the wafers to annealing, it is possible to establish ohmic connections on both n-InP and p-InGaAs [99]. The photonic devices were created using a unique architecture of III-V epitaxial layers, where the laser and photo-detector heterostructures are stacked together in the same epitaxy [100]. This enables a far easier integration method and dramatically reduces the size of optical communications compared to previously established point-to-point connections, where specific dice are allocated for laser structures and others for photo-detector structures. The utilization of 248 nm deep ultraviolet (DUV) lithography is feasible because of the 1 μm overall epitaxial thickness, which eliminates issues related to focusing. The given process, known as III-V, is optimized for a 200 mm wafer size. It involves wet and $\text{CH}_4\text{-H}_2$ -based reactive ion etching, deposition of an oxide isolation layer, and metallization [101]. Fig. 9 illustrates an optical microscope image showing a 130 nm thick SiO_2 bonding layer that covers a waveguide, an III-V microdisk laser, and a photo-detector. The electrical connections did not include the contact area and the usual gold-based metallization. Alternatively, a metal stack consisting of Ti/TiN/AlCu is applied uniformly throughout the entire wafer. After lithography, a chlorine-based chemical is employed to perform dry-etching on the metal stack till reaching the oxide isolation layer, which also functions as an etch-stop layer. By not subjecting the wafers to annealing, it is possible to establish ohmic connections on both n-InP and p-InGaAs.

3. Limitations and future scope

Moreover, silicon presents certain challenges in its application as an optical or photonic material. These problems are basic restrictions resulting from material characteristics. The challenge of externally controlling silicon structures for optical modulation and switching arises from the nonlinear coefficient of silicon and Si-based electro-optic (EO) switches, which require high drive powers. Additionally, the lack of an all-silicon optical source forces coupling between nano-scale Si optical structures and micron-sized fibers. Significant losses are caused by the

high refractive index and mode mismatch between the silicon waveguides and fibers, which in turn makes coupling challenging. Light confinement can raise the electromagnetic field in certain areas of the structure by causing an enhancement of electro-optical and nonlinear phenomena in silicon. Strong light confinement may also be used to control the optical mode by varying the silicon waveguide's characteristics. The present state of the art is reviewed together with light coupling, switching, and modulating on Si. Unlike previous studies, these results are based on very restricted submicron structures. This technique, which deals with confinement-based light manipulation, has primarily enabled the use of ultra-compact structures on silicon.

The twentieth century saw the development of microelectronics and photonics, which revolutionized engineering, information science, and telecommunications. It is hard to imagine any branch of engineering in the contemporary era that has influenced our way of life and work in such a profound, all-encompassing, and positive way. Thanks to developments in microelectronics, optoelectronics, and photonics, we are now able to enjoy incredible advances in science and technology that are extraordinary in both extent and sophistication. These “high technology” areas in today's world control everything, from a washing machine to a space shuttle, and are equally important from the standpoint of an air force pilot, a young video enthusiast, a banking executive, an entertainer, or a housewife. The remarkable growth of the Internet, wherein its user base doubles every few months, serves as an excellent illustration. Microelectronics and photonics have profound implications for business, jobs, national security, and maybe even how society is organized in the future. The development of intelligent instruments, goods, and procedures is essential to the industrialized world's survival. It makes sense for everyone, not just experts, to be curious about potential new applications for photonics and microelectronics that may appear in the near future and what kind of influence they will have in the twenty-first century.

4. Conclusion

Major technological advances are being made in integrating photonic and microelectronic components and circuits on a single bulk silicon chip. These exciting technological advances are happening at a time when there is an ever-increasing demand for higher-performance optoelectronic technologies that are expected to change the landscape and our daily lives. We are gradually moving from an expensive microelectronic fabrication process to an era of lower-cost processes that yield higher-performing and low-power-consuming technologies. High data rates, deep integration, and optical functionalities have shown silicon

photonics to be a formidable foundation for a wide range of communication applications. Despite the above-mentioned advancements, several areas still require work to scale silicon photonic device integration. We list a few obstacles to silicon photonic technologies, such as power loss, which includes the loss due to passive components, delay lines, switches, and chip interconnects, which is a major problem for integrated photonic circuits. The next generation of ultralow-loss components could be made possible via multimode waveguide technology and mixing with silicon or silicon nitride. Additionally, an integrated silicon optical chip is temperature-sensitive; thus, temperature significantly impacts the grating structure used to design optical devices. Therefore, to avoid this influence of temperature, simulation design, and structural design optimization are required. Moreover, these electronic-photonic chips are expected to revolutionize circuit design and enable a wide range of new applications that include high bandwidth, and low power interconnects within chips and highly sensitive biological/chemical sensors and photonic chips for RF signal transmission and processing in cellular networks and radar, just to name a few examples of applications. More technological breakthroughs are anticipated shortly in integrating photonics and microelectronics for various new technological applications. The future of integrated microelectronics and photonics looks bright for systems-on-a-chip evolution.

Declaration of competing interest

The authors declare that they have no known competing financial interests.

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